

AI StudyBuddy

AI-Powered Study Management and Learning Support System

PHASE 2

Requirement Analysis

Role	Team Member
Team Leader	Kishore T
Team Member	Logeshwaran D
Team Member	Manikandan M
Team Member	Mohamed Safir M

Submitted as a phase-wise project submission.

1. Functional Requirements

- Register a student or administrator account.
- Authenticate registered users and issue access/refresh tokens.
- Upload study material.
- Generate AI-based summaries, flashcards, quizzes and study plans.
- Allow administrators to retrieve registered users.
- Allow administrators to delete a user and associated materials.

2. Non-Functional Requirements

- Authentication and protected routes should prevent unauthorized administrator operations.
- The backend should provide clear HTTP responses for successful and failed requests.
- The application should be maintainable through modular controllers, routes, middleware, models and utilities.
- Sensitive configuration should be kept in environment variables.

3. Technology Requirements

Technology/Tool	Purpose
Node.js	Runtime environment
Express.js	Backend framework
MongoDB	Database
Mongoose	MongoDB object modeling
JWT	Access and refresh token authentication
Google Gemini API	AI integration
Visual Studio Code	Development environment
Postman / REST client	API testing
Nodemon	Development server utility

4. Environment Configuration

The project uses environment variables for the server port, MongoDB connection, JWT secrets, Gemini API key and development environment. Actual secrets are not included in this submission.